

Ben Bubnick

DATA ENGINEER · DATA SCIENTIST · ANALYTICS SPECIALIST

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Summary

Experienced Data Engineer with a strong track record of delivering scalable health data pipelines across cloud platforms, automating validation processes, and documenting workflows.

Experience

GrowData Analytics

AWS: S3, Glue, Athena, Lambda, CloudWatch, Ruby, RubyLLM, Tesseract OCR, Rubocop, RSpec

PRINCIPAL ML ENGINEER

Sep. 2024 - Present

Built and shipped an OCR-to-analytics product for small farms; owned model integration, CI/CD, testing, and auditability on UNIX-like systems.

- Productized OCR → structured data pipeline for farm invoices (handwritten/printed) using Tesseract + rule-based post-processing; shipped as a small Python/Ruby service with containerized deployments and versioned configs.
- Built validation & reconciliation layers (schema checks, cross-field rules, idempotent re-runs) and produced audit logs to support bookkeeping/compliance workflows.
- Implemented CI/CD (unit tests, linting, fixtures, golden-file tests) and release automation; documented runbooks and failure modes (timeouts, low-confidence OCR, retries).
- Delivered a lightweight analytics layer (usage metrics, error taxonomy, latency/throughput dashboards) to guide feature tuning and cost control.
- Owned infra as code for a minimal AWS footprint (S3 for artifacts, Lambda/Cron for jobs, CloudWatch alerts); operated via UNIX-like CLI workflows (Bash/Zsh, Makefile targets).

Health Data Movers

AWS: S3, Snowflake, SQLMesh, Python, SQL, Docker

DATA CONSULTANT

Mar. 2025 - Present

Developed a reusable ETL pipeline to streamline data onboarding for a healthcare revenue cycle platform.

- Built a reusable ingestion & transformation framework in SQLMesh/Snowflake (templated models, data contracts, CI checks, promotion gates), enabling consistent, repeatable onboardings.
- Added observability & auditability (row-count/parity checks, schema drift alerts, reconciliation logs) and wrote runbooks for deterministic re-runs and incident handling on UNIX-like systems.
- Shipped a lightweight forecasting/alerting pattern (Python service + Docker + Actions) for operations signals; packaged with tests and deployment automation.

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

SENIOR DATA ANALYST

Jun. 2023 - Jan. 2025

Built a scalable AWS data lake with automated CI/CD, metadata reporting, and integrated analytics tools.

- Stood up an AWS data lake with CI/CD (dbt + Git flows), cost-aware partitioning, and job scheduling; reduced deployment time 50% and improved query performance with targeted model refactors.
- Implemented monitoring/alerting & on-call runbooks (metrics, logs, failure taxonomies) to lower MTTR and improve data reliability.
- Mentored engineers on code reviews, testing, and pipeline design; authored internal standards that lifted team delivery quality.

Merative

Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps

SENIOR DATA SCIENTIST

Jul. 2022 - Mar. 2023

Launched a new Azure platform to replace HDFS systems to modernize data infrastructure.

- Led HIPAA-compliant deidentification initiative across 200+ million patient records.
- Designed and implemented an automated testing framework for ETL pipelines for validation coverage.
- Validated and optimized Azure-based data workflows, ensuring continuity and fidelity during the transition from legacy HDFS systems to cloud architecture.
- Conducted end-to-end system testing for Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations.
- Collaborated with engineering teams across disciplines to refine architectural designs.
- Documented test plans and platform migration procedures for improving onboarding for future QA engineers.
- Built regression test suites and data comparison tools, detecting schema drift and transformation errors early in the development lifecycle.
- Ensured compliance with data governance and security protocols to reduce audit risks and support HITRUST readiness.

IBM

Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle

SENIOR DATA SCIENTIST

Nov. 2021 - Jul. 2022

Built AI tools while leading analytics delivery and large-scale data validation efforts.

- Created AI Surgical Scheduling Optimization tool currently evolving into new offering.
- Led delivery on tailored analytics and reporting project for client administrative data.
- Developed data migration validation analysis on 150+ billion patient records.
- Developed surgical smart case card using combined CNN/RNN model.

SENIOR DATA CURATION ARCHITECT

Dec. 2020 - Nov. 2021

Led large-scale clinical data and NLP initiatives and delivering ML solutions.

- Developed clinical data integration of 100+ billion structured & semi-structured records.
- Developed topic modelling project on nearly 6 million semi-structured provider notes data.
- Curated 25 pages of workflow documentation and mentored new hires in AI tuning process.
- Created COVID-19 hospitalization prediction tool with Machine Learning algorithms.

DATA CURATION ARCHITECT

Nov. 2017 - Dec. 2020

Led enterprise data integration initiatives and mentoring programs.

- Managed multi-org enterprise integration with over 70 interconnected data sources.
- Developed new data validation and review process that reduced total rework by 25%.
- Reduced mapping rework by 45% with through new knowledge-driven initiatives.
- Led mentoring program to increase 4,500 billable hours over two year period.

SOFTWARE DEVELOPER

Sep. 2016 - Nov. 2017

Developed internal resources that reduced costs and improved onboarding efficiency.

- Reduced time spent on knowledge transfer by 80% by rewriting project documentation.
- Eliminated need for a \$30,000 annual contract by developing data curation tool.
- Reduced cost-per-hire by \$5,210 by re-developing education program.

DATA SCIENTIST

Nov. 2015 - Sep. 2016

Led integration efforts and process automation initiatives.

- Managed team of 12 remote data science contractors for 36 integration projects.
- Created over 125 new knowledge base articles, making up 30% of the team's online content.
- Developed automated data integration tool to cut out 28% unnecessary development time.

Explorys

Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENTIST

Jun. 2015 - Nov. 2015

Built ETL pipelines on distributed data systems to support clinical data integration and population analytics.

- Developed cron-based widget to define blackout periods for client data pulls.
- Performed troubleshooting on production issues across multiple environments.
- Principal integration specialist for 10+ client database warehouses.

Multiple Institutions

ADJUNCT PROFESSOR

Aug. 2010 - May. 2019

Taught physics & astronomy lecture series and lab courses.

- Consistently exceeded student evaluation benchmarks, surpassing institutional feedback goals by an average of 16% in 2017 and 17% in 2018, reflecting strong teaching effectiveness and student satisfaction.
- Improved student comprehension and course navigation by reorganizing online instructional materials, resulting in a 28% reduction in reported confusion and fewer clarification requests during office hours.
- Developed and published over 700 hours of recorded lectures and lab content, enhancing accessibility and supporting flexible, on-demand learning for both traditional and non-traditional students.
- Implemented Just-in-Time Teaching (JiTT) strategies for underperforming students, leading to an average grade improvement of 18% by the end of term.
- Designed interactive lab lectures based on the Interactive Lecture Demonstration (ILD) method, increasing student engagement and concept retention in physics instruction.
- Promoted collaborative learning through Peer Instruction and ILD techniques, effectively addressing real-time student misconceptions during physics lab sessions.

AmTrust Financial Services

VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL

SOFTWARE ENGINEER

Mar. 2014 - Jun. 2015

Developed MVC components and web forms for insurance policy software.

- Led development on core insurance policy software, serving as principal developer for a new product offering and contributing to two major architectural initiatives in 2015.
- Analyzed and optimized performance across 65+ ISO rating-based web forms using R, identifying bottlenecks and reducing page load times.
- Championed and implemented a unit testing framework for multi-tier MVC components, improving code quality and reducing deployment errors across the dev team.

PPG Industries

VB.NET, C#, SQL, C++, FORTRAN

COLOR SCIENTIST

Apr. 2012 - Jun. 2013

Developed MVC components and web forms for insurance policy software.

- Improved color matching classification algorithm accuracy by 40% by developing and applying advanced numerical methods, enhancing product performance across automotive coatings.
- Developed cross-instrument correlation software to support a risk analysis initiative, enabling consistent color measurement across varying hardware platforms and improving QA reliability.
- Supported the creation of an industry-leading automated color matching facility, developing numerical tools and methods to improve formulation quality and pigment data analysis in high-variance OEM color environments.

Awards

2018 - 2022	Gold Champion , For 1,477 classroom hours and earning 43 digital credentials	IBM
2018	Lab Services Award , For coordinating internal teams to deliver solutions on time	IBM
2017	Eminence and Excellence Award , For developing a new data curation tool	IBM
2012	Color Lab Automation Award , For expanding data collection and analysis reports	PPG Industries
2010	Mary J. Hanna Fellowship , For STEM students pursuing research-based Masters	UC

Volunteer

HSEA

ETL BRIGADE

Feb. 2025 - May. 2025

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

CSforAll

INSTRUCTOR

Oct. 2018 - May. 2019

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE

COACH

Mar. 2017 - Mar. 2019

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America

ETL GUILD

Jan. 2015 - Jun. 2015

The initiative was brought about to apply data science tools to improve and expand public record access.

T.C.P. World Academy

TUTOR

Sep. 2008 - May. 2009

Our goal was to teach students how to understand the value of their learning experience and becoming academically involved by participating in society simulation activities for higher learning.

Talks

AI Hackathon 2020: Surgery Scheduling Optimization

"An algorithm uses past utilization and current waitlist info to predict surgery duration.

Dec. 2020

IBM

REACH Initiative: IBM Watson Care Insights

Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow.

2019

IBM

Content & Data Analytics Learning Exchange

Introduction of implementation of machine learning algorithms using real-world health data analytics.

2018

IBM

2018 Innovation Research Interchange Fall Networks Conference

Using NLP to extract insight from unstructured data and transform it from a data lake.

Sep. 2018

IRI

Cross Team Training Series

A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes.

2016

EXPLORYS

Publications

Reproducible Postgres load of Tuva seed datasets

Reproducible Postgres load of Tuva seed datasets with robust validation.

Aug. 2025

HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction ^a

Improvement on traditional fixed ratio methods for total procedure time estimation.

Dec. 2020

IBM

Process and Requirements for CCD Data Integration

Processes and requirements for overcoming the many difficulties with CCD data curation.

Oct. 2017

IBM

AI Paint Formula Prediction from Spectroscopic Measurements ^b

Naive bayesian color and effects matching algorithm of complex automotive paint formula.

May. 2015

SHINYAPPS.IO

Characterization of Surface Coatings using RT Models for Color Matching

Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching.

Feb. 2013

USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy ^c

Spectroscopic and Photometric analysis of two open clusters in the galactic plane.

Dec. 2010

OHIOLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107

Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX.

May. 2008

VIZIER

**Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster
[BDS2003] 107**

Feb. 2008

New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107).

PASP

A Novel Method for Low-Toxic Photo-Etch Printing^d

May. 2003

Application of aquatint etching techniques to photo-etch procedures.

MAPC

Education

2022 **PGCert.**, Data Science Professional^a

IBM

2015 **PGCert.**, Data Science^b

Johns Hopkins

University

2010 **Master of Science**, Astrophysics^c

University of Cincinnati

2004 **Bachelor of Fine Arts**, Printmaking^d

Ohio University